

Exercise Sheet 12b - Solutions

Predicate Logic – Natural Deduction & Semantics

1. Here is a constructive Natural Deduction proof of $(S_1) \rightarrow (S_2) \rightarrow \perp$

$$\begin{array}{c}
 \frac{\overline{S_1}^1 \quad \frac{\overline{\forall y.x < y}^3 \quad \frac{x < x}{\perp} [\forall E] \quad \frac{\overline{S_2}^2 \quad \neg x < x}{\perp} [\forall E] [\neg E]}{\perp}^3 [\exists E]}{\frac{\perp}{(S_2) \rightarrow \perp}^2 [\rightarrow I]}^1 [\rightarrow I] \\
 (S_1) \rightarrow (S_2) \rightarrow \perp
 \end{array}$$

2. Here is a constructive Natural Deduction proof of $(S_2) \rightarrow (S_3) \rightarrow \forall x.x \leq x$

$$\begin{array}{c}
 \frac{\overline{S_3}^2 \quad \frac{\forall y.x < y \vee y \leq x}{x < x \vee x \leq x} [\forall E]}{\frac{\overline{x < x}^3 \quad \frac{\overline{S_2}^1 \quad \neg x < x}{\perp} [\forall E] [\neg E]}{x \leq x} [\perp E]}^3 [\rightarrow I] \quad \frac{x \leq x}{x \leq x \rightarrow x \leq x}^4 [\rightarrow I]}{\frac{x \leq x}{\forall x.x \leq x} [\forall I]}^2 [\rightarrow I] \\
 (S_2) \rightarrow (S_3) \rightarrow \forall x.x \leq x
 \end{array}$$

3. For example, the following models are models of S_3 :

- $M_1 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n + 1 \rangle, \langle \{ \langle n, m \rangle \mid n < m \}, \{ \langle n, m \rangle \mid n \leq m \} \rangle \rangle$
- $M'_1 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n + 1 \rangle, \langle \{ \langle n, m \rangle \mid n \leq m \}, \{ \langle n, m \rangle \mid n < m \} \rangle \rangle$
- $M''_1 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \{ \langle n, m \rangle \mid \text{True} \}, \emptyset \rangle \rangle$

4. For example, the following models are not models of S_3 , i.e., are models of $\neg S_3$:

- $M_2 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n + 1 \rangle, \langle \{ \langle n, m \rangle \mid n < m \}, \{ \langle n, m \rangle \mid n < m \} \rangle \rangle$
- $M'_2 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \{ \langle n, m \rangle \mid n < m \}, \emptyset \rangle \rangle$
- $M''_2 = \langle \mathbb{N}, \langle 0, \langle n \rangle \mapsto n \rangle, \langle \emptyset, \emptyset \rangle \rangle$